

# Module 08: Planning in High School

## Learning Objectives

1. Define **Planning** within the context of High School.
2. Analyze how Planning interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Planning.

## Introduction

This module explores **Planning**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Planning is a critical component of the 8-part Active Inference spine, bridging the gap between Communication and Systems.

## Key Concepts

### 1. Planning as a Markov Blanket Boundary

How does Planning define the boundary between the agent and the environment?

### 2. Generative Models of Planning

What parameters involved in Planning must be optimized to minimize variational free energy?

### 3. Active Inference Dynamics

How does the process of Planning drive the perception-action loop?

## Applications

In High School, we see Planning manifest in: \* **Specific Example 1:** Google Maps route planning is Active Inference in a technology system: the app maintains a generative model of

traffic conditions built from millions of phone GPS signals, simulates multiple possible routes with their predicted travel times, and recommends the path that minimizes expected free energy (arrival time uncertainty plus delay risk) -- dynamically re-planning when a traffic jam creates a prediction error on your current route. \* **Specific Example 2:** An AI game agent like AlphaGo performs planning through Monte Carlo Tree Search, which is Active Inference over future game states: it simulates thousands of possible move sequences (policies), evaluates each terminal state's expected outcome, and selects the move that minimizes expected free energy across the entire game tree -- balancing the exploitation of known winning patterns with exploration of novel strategies the opponent might not expect.

## Conclusion

Understanding Planning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.